CS 3510 Algorithms: Linear Programming

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Linear Programming Problems

- A set of variables to find assignments for: $x_1, ..., x_n$
- A linear function to maximize: $f(x_1, ..., x_n) = c_1 x_1 + ... + c_n x_n$
- A set of linear inequalities of the form: $\vec{a}_i \cdot \vec{x} \leq b$

Any vector (x_0) that satisfies the inequalities is said to be *feasible*.

We are looking for the optimal solution (x^*) that is feasible and maximizes f.

Example: Cheapest Healthy Diet

A kg of each food has nutritional content and a price:

Food	**Calories**	**Calcium**	**Magnesium**	**Vitamin C**	**Price**
Milk	3000	32	720	10	12.00
Liver	1000	12	1200	20	2.60
Orange	150	2	100	100	9.50
Beef	1400	9	210	0	11.50
Spinach	220	40	2000	60	11.50

A human has daily requirements: 2000 calories, 22 of calcium, 120 of magnesium, 15 of vitamin C.

What is the cheapest healthy diet?

Example: Cheapest Healthy Diet

Let $x_1, \ldots, x_5$ be the amount of each food in the daily diet.

We are minimizing $f(\vec{x}) = c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4 + c_5x_5$

There is a contraint for each dietary requirement:

$a_{i,1}x_1 + a_{i,2}x_2 + a_{i,3}x_3 + a_{i,4}x_4 + a_{i,5}x_5 \geq b_i)$

And you can't have negative amounts of food:

$x_i \geq 0$ for all i

An example we can draw

Item 1 sells for \$1, Item 2 sells for \$6.

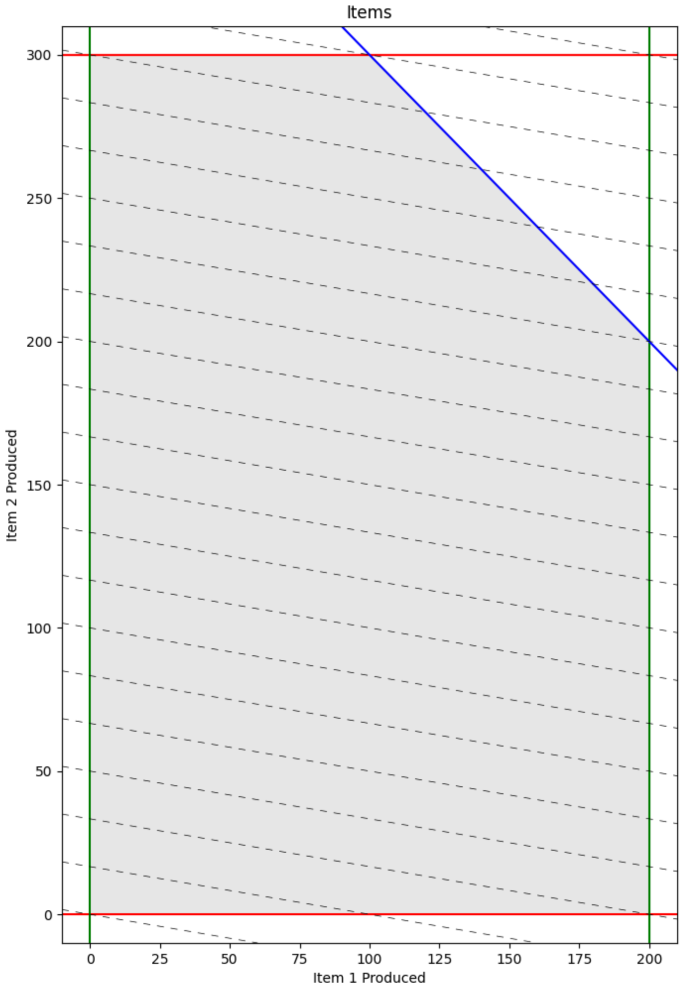
Maximize: $1.0x_1 + 6.0x_2$

Each day, factory can only produce:

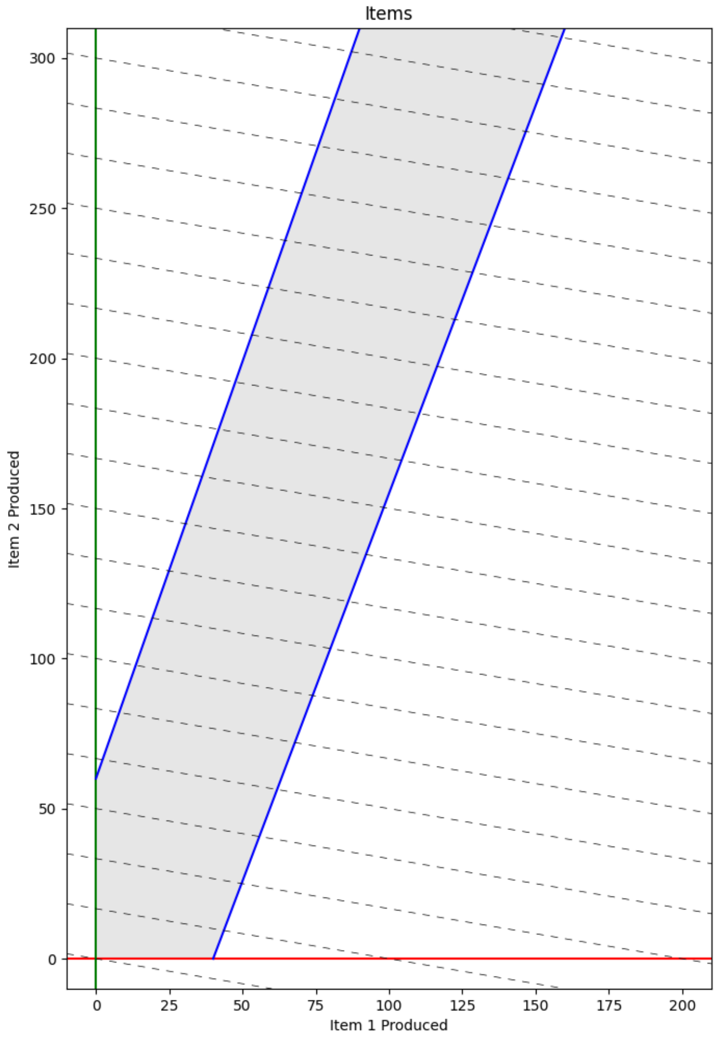
- 400 or less total items
- No more than 200 of Item 1
- No more than 300 of Item 2

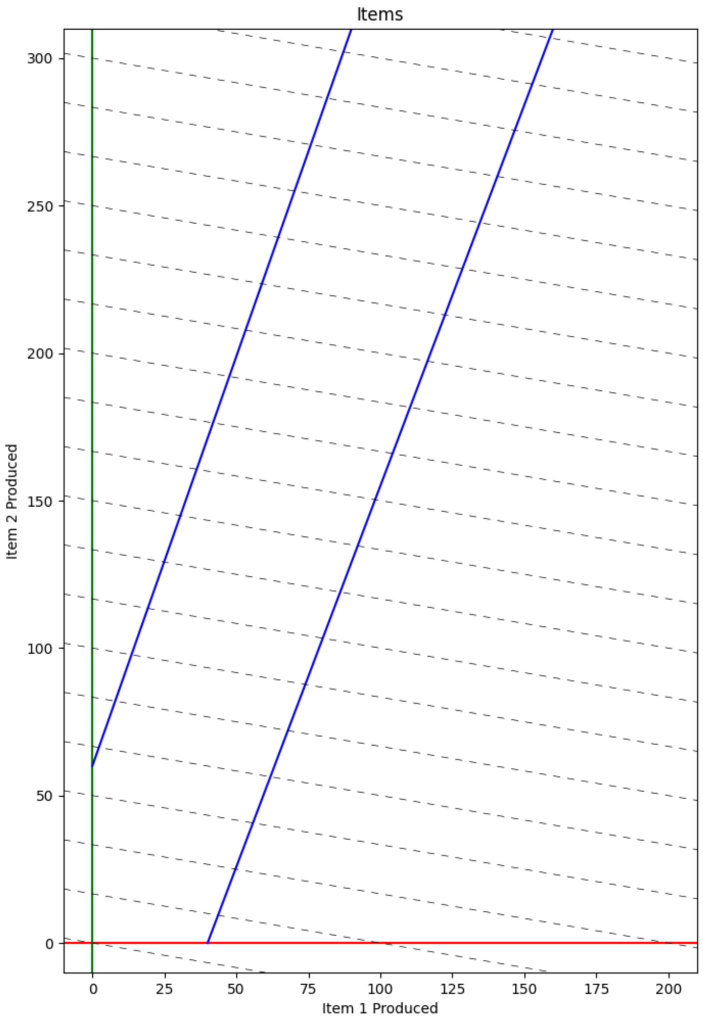
Constraints:

- $x_1 \leq 200$
- $x_2 \leq 300$
- $x_1 + x_2 \leq 400$
- $x_1 \geq 0$, $x_2 \geq 0$



Unsolveable





We Love Matrices

Maximize: $f(\vec{x}) = 1.0x_1 + 6.0x_2$

Constraints:

- $g_1(\vec{x}) = 1.0x_1 + 0.0x_2 \leq 200$
- $g_2(\vec{x}) = 0.0x_1 + 1.0x_2 \leq 300$
- $g_3(\vec{x}) = 1.0x_1 + 1.0x_2 \leq 400$
- $x_1 \geq 0,\, x_2 \geq 0$

Variables as a vector :

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\vec{c} = \begin{pmatrix} 1.0 \\ 6.0 \end{pmatrix} = \nabla f$$

Maximize: $\vec{c}^T \vec{x}$

$$\boldsymbol{A} = \begin{pmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \\ 1.0 & 1.0 \end{pmatrix} = \begin{pmatrix} \nabla g_1 \\ \nabla g_2 \\ \nabla g_3 \end{pmatrix}$$

$$\vec{b} = \begin{pmatrix} 200.0 \\ 300.0 \\ 400.0 \end{pmatrix}$$

Constraints:

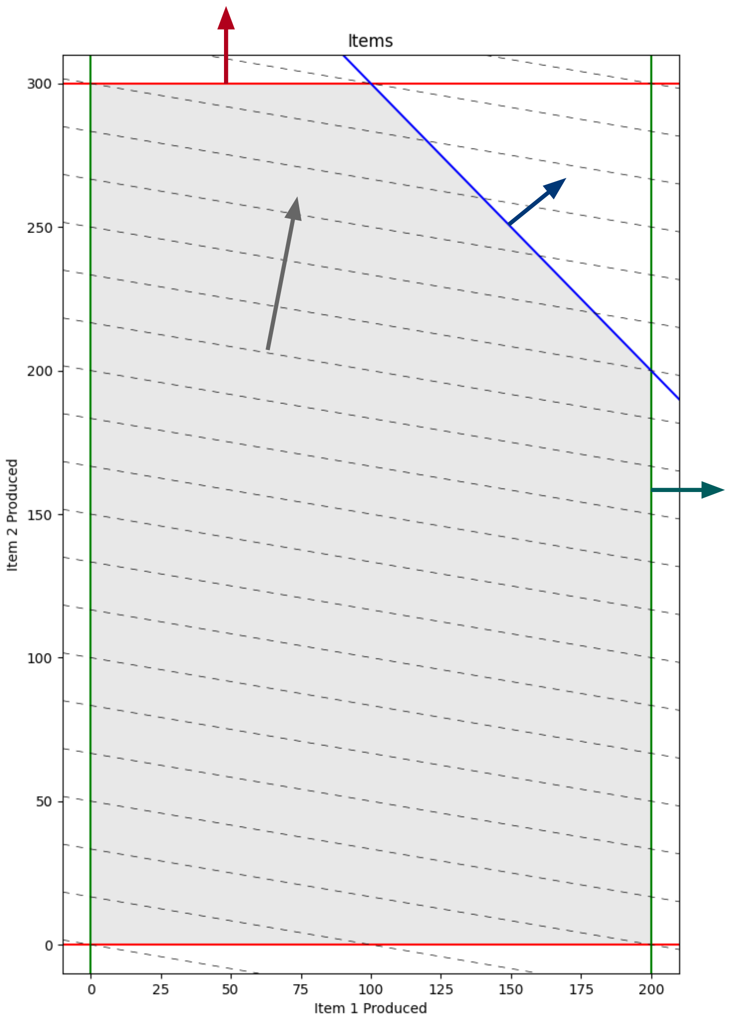
- $\boldsymbol{A}\vec{x} \leq \vec{b}$
- $\vec{x} \geq 0$

Think Gradients

Maximize: $f(\vec{x}) = 1.0x_1 + 6.0x_2$

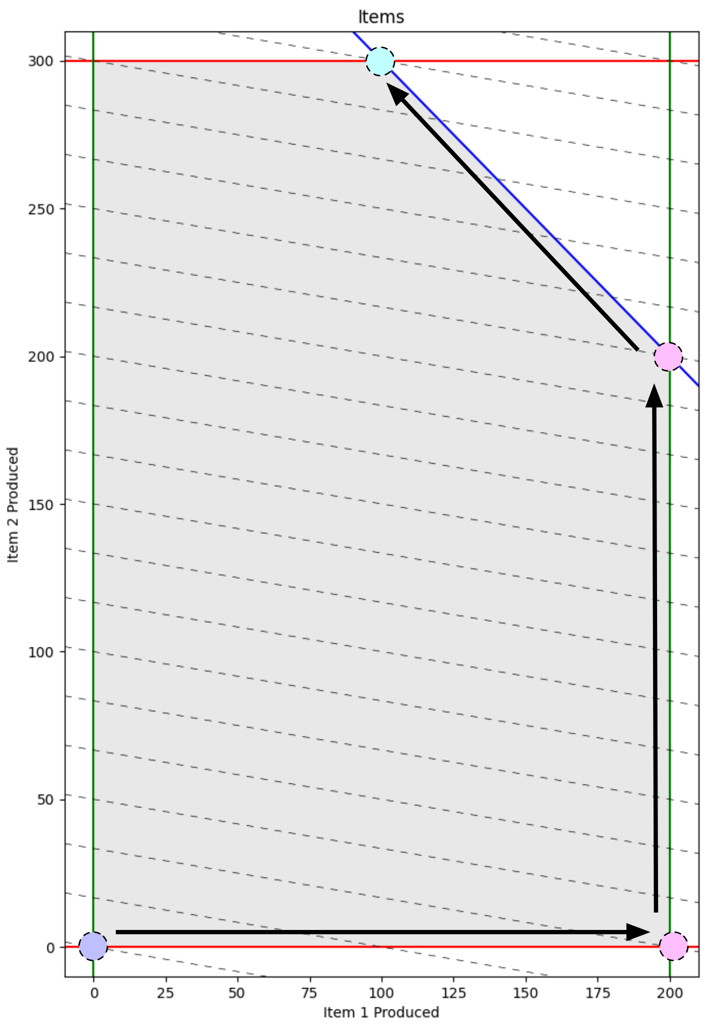
Constraints:

- $g_1(\vec{x}) = 1.0x_1 + 0.0x_2 \leq 200$
- $g_2(\vec{x}) = 0.0x_1 + 1.0x_2 \leq 300$
- $g_3(\vec{x}) = 1.0x_1 + 1.0x_2 \leq 400$



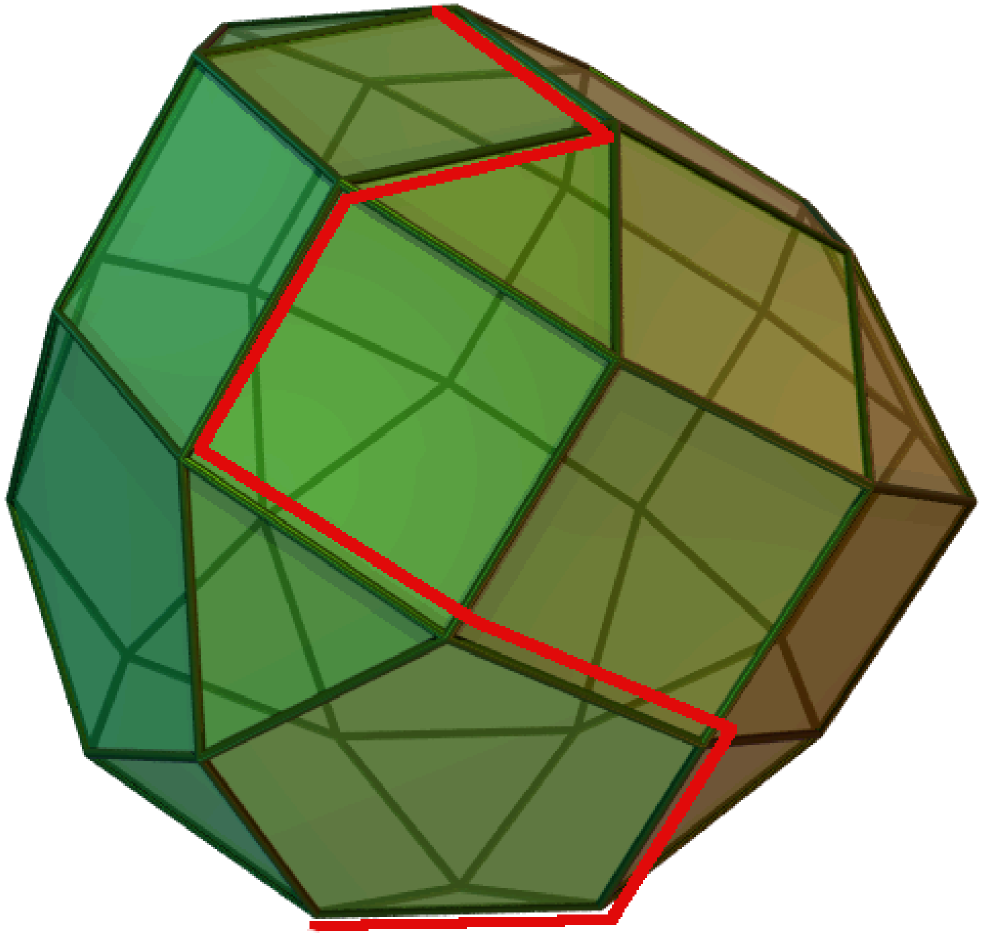
The Simplex Method

- Start at any corner of the feasible region
- Walk an edge uphill until you hit another corner
- Repeat until none of the edges you are on go uphill



With three variables

- Level sets of g and f are planes
- Inequalities define half-spaces
- Feasible region is convex 3D polytope
- If bounded and non-empty, an optimal solution definitely exists.
- And at least one corner is an optimal solution.



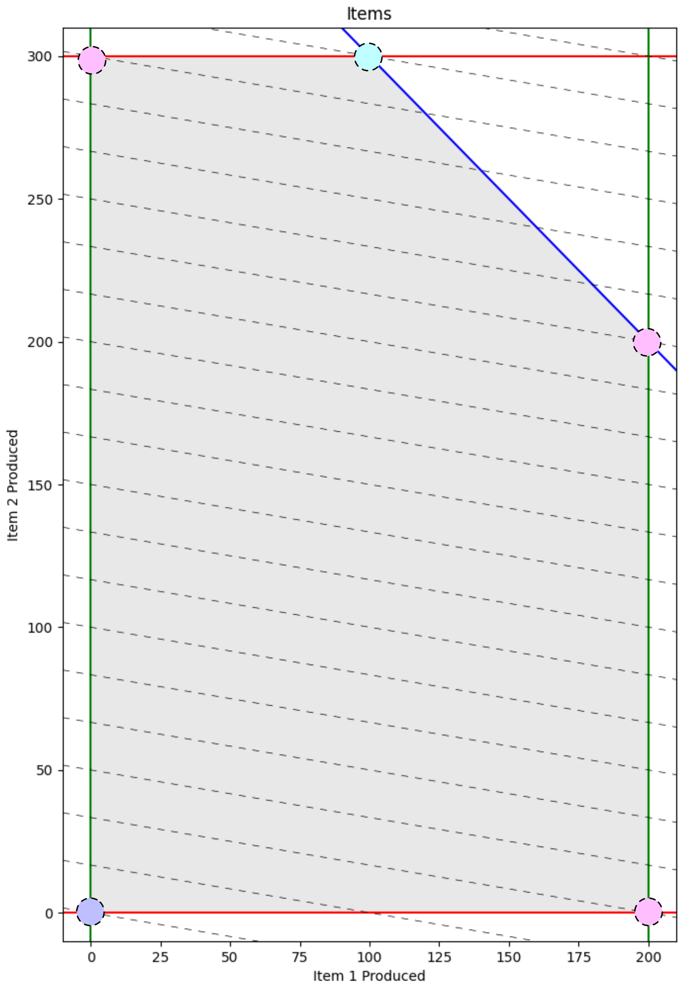
Slack Variables

How do we define "the corners"?

- $1.0x_1 + 0.0x_2 + s_1 = 200$
- $0.0x_1 + 1.0x_2 + s_2 = 300$
- $1.0x_1 + 1.0x_2 + s_3 = 400$
- $x_1, x_2, s_1, s_2, s_3 \geq 0$

At each corner, how many are zero?

How many real variables are we solving for?



Implementing Simplex

Often $\vec{x} = \vec{0}$ is feasible. An easy place to start: Only slack variables are non-zero.

Variables that are non-zero are “basic”.

At each corner, we pick an “entering” variable (no longer needs to be zero) and an “exiting” variable (one that will be locked down as zero). Then we maximize f by solving for values for all “in” variables.

There are two methods for implementing the simplex algorithmF:

- Tableau – row operations, makes perfect sense if you are solving by hand.
- Revised – vector operations, makes it easy to implement on a computer.

Computation Complexity

Let n be the number of variables.

Worst case is exponential in n.

But, in practice, it is surprisingly efficient: polynomial in n.

Quick Example

- Maximize: $2x_1 + 3x_2 + 5x_3$
- Subject to:
  - $x_1 + 2x_2 + x_3 \leq 4$
  - $x_1, x_2, x_3 \geq 0$

Linear Programming is Kinda Easy

- Linear Integer Programming seems easier, but...

Tableau Example

- Maximize: $z = 3x_1 + x_2 + 2x_3$
- Subject to:
  - $x_1 + x_2 + 3x_3 \leq 30$
  - $2x_1 + 2x_2 + 5x_3 \leq 24$
  - $4x_1 + x_2 + 2x_3 \leq 36$
  - $x_1, x_2, x_3 \geq 0$
- Rewrite with slack variables:
  - $s_4 = 30 - x_1 - x_2 - 3x_3$
  - $s_5 = 24 - 2x_1 - 2x_2 - 5x_3$
  - $s_6 = 36 - 4x_1 - x_2 - 2x_3$
- Start search at $\vec{x} = \vec{0}$
- Increasing which variable will most increase z?
- Which constraint limits increase of x_1?
- x_1 enters, s_6 exits:
  - $x_1 = 9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{s_6}{4}$
- Rewrite other equations:
  - $z = 27 + \frac{x_2}{4} + \frac{x_3}{2} - 3\frac{s_6}{4}$
  - $s_4 = 21 - \frac{3x_2}{4} - \frac{5x_3}{2} + \frac{s_6}{4}$
  - $s_5 = 6 - \frac{3x_2}{2} - 4x_3 + \frac{s_6}{2}$
- Which variable should we let be non-zero now?
- Which equality limits x_3?
  - $x_3 = \frac{3}{2} - \frac{3x_2}{8} - \frac{s_5}{4} + \frac{s_6}{8}$
- Rewrite other equations and repeat:
  - $x_1 = \frac{33}{4} - \frac{x_2}{16} + \frac{s_5}{8} - \frac{5s_6}{16}$
  - $s_4 = \frac{69}{4} + \frac{3x_2}{16} + \frac{5s_5}{8} + \frac{s_6}{16}$
  - $z = \frac{111}{4} + \frac{x_2}{16} - \frac{s_5}{8} - \frac{11s_6}{16}$

Simplex?

- We are walking corners:
  - Started $(0,0,0,30,24,36) \to z = 0$
  - Moved to $(9,0,0,21,6,0) \to z = 27$
  - Moved to $\left(\frac{33}{4}, 0, \frac{3}{2}, \frac{69}{4}, 0, 0\right) \to z = 27.75$
- What variable do we want to allow to be non-zero?
  - $z = \frac{111}{4} + \frac{x_2}{16} - \frac{s_5}{8} - \frac{11s_6}{16}$
- Which equation limits x_2?
  - $x_1 = \frac{33}{4} - \frac{x_2}{16} + \frac{s_5}{8} - \frac{5s_6}{16}$
  - $x_3 = \frac{3}{2} - \frac{3x_2}{8} - \frac{s_5}{4} + \frac{s_6}{8}$
  - $s_4 = \frac{69}{4} + \frac{3x_2}{16} + \frac{5s_5}{8} + \frac{s_6}{16}$
- $x_2 = 4 - \frac{8x_3}{3} - \frac{2s_5}{3} + \frac{s_6}{3}$
- Rewrite other equations:
  - $z = 28 - \frac{x_3}{6} - \frac{s_5}{6} - \frac{2s_6}{3}$
  - $x_1 = 8 + \frac{x_3}{6} + \frac{s_5}{6} - \frac{s_6}{3}$
  - $x_4 = 18 - \frac{x_3}{2} + \frac{s_5}{2}$
- No way to increase z. Final solution:
  - $(8,4,0,18,0,0) \to z = 28$

Primal vs. Dual

Primal Problem

Maximize $\vec{c}^T \vec{x}$

Subject to $\boldsymbol{A}\vec{x} \leq \vec{b}$ and $x \geq 0$

From our example:

$$\vec{c}^T \vec{x} = [3, 1, 2] \begin{pmatrix} 8 \\ 4 \\ 0 \end{pmatrix} = 28$$

Dual Problem

Minimize $\vec{b}^T \vec{y}$

Subject to $\boldsymbol{A}^T \vec{y} \geq \vec{c}$ and $\vec{y} \geq 0$

Also our example:

$$\vec{b}^T \vec{y} = [30, 24, 36] \begin{pmatrix} 0 \\ \frac{1}{6} \\ \frac{2}{3} \end{pmatrix} = 28$$

$$\max\ \vec{c}^T \vec{x} = \min\ \vec{b}^T \vec{y}$$

Proof of Weak Duality

Show that: $\max\ \vec{c}^T\vec{x} \leq \min\ \vec{b}^T\vec{y}$

$$\max\ \vec{c}^T\vec{x} = \vec{x}^T\vec{c} \leq \vec{x}^T\boldsymbol{A}^T\vec{y} = (\boldsymbol{A}\vec{x})^T\vec{y} \leq \vec{b}^T\vec{y}$$

Simplex Solves Primal and Dual

Minimize $30y_1 + 24y_2 + 36y_3$

Subject to:

- $y_1 + 2y_2 + 4y_3 \geq 3$
- $y_1 + 2y_2 + y_3 \geq 1$
- $3y_1 + 5y_2 + 2y_3 \geq 2$
- $y_1, y_2, y_3 \geq 0$

Reminder:

- $z = 28 - \frac{x_3}{6} - \frac{s_5}{6} - \frac{2s_6}{3}$
- s_4 is basic, so $y_1 = 0$
- Coefficent of $s_5 = -\frac{1}{6}$, so $y_2 = \frac{1}{6}$
- Coefficent of $s_6 = -\frac{2}{3}$, so $y_3 = \frac{2}{3}$

$30(0) + 24\left(\frac{1}{6}\right) + 36\left(\frac{2}{3}\right) = 28$

Questions?

Slides by Aaron Hillegass

Based on lectures by Abrahim Ladha